MA.7.GR.1.2

Benchmark

MA.7.GR.1.2	**Solve mathematical or real-world problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals.**

Benchmark Clarifications:
Clarification 1: Within this benchmark, the expectation is not to find areas of figures on the coordinate plane or to find missing dimensions.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.2.1, MA.7.NSO.2.2, MA.7.NSO.2.3
- MA.7.GR.2.2

Terms from the K-12 Glossary

- Area
- Composite Figure
- Polygon
- Quadrilateral
- Triangle

Vertical Alignment

Previous Benchmarks

- MA.6.GR.2.2

Next Benchmarks

- MA.912.GR.3.4
- MA.912.GR.4.3, MA.912.GR.4.4

Purpose and Instructional Strategies

In grade 6, students solved problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles. In grade 7, students solve problems involving the area of polygons or composite figures by decomposing them into triangles or quadrilaterals. In high school, students will extend this knowledge to solve mathematical and real-world problems involving the perimeter or area of any polygon using coordinate geometry and other tools.

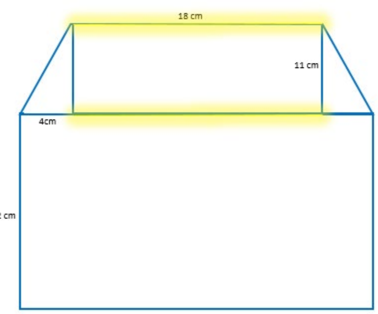
- Instruction includes problems where multiple decompositions are possible so students understand the various pathways to a solution (*MTR.5.1*). Scaffolded instruction may include figures on grid paper to allow students to more easily count the total area. Select and order student solutions to be shared with the whole group (*MTR.4.1*), depicting various solution pathways.
- Instruction includes figures where an efficient method is to subtract a basic figure from a larger figure.
    - Students should use grid paper to draw a polygon that is composed of triangles and quadrilaterals that can be exchanged with a partner or within a group to find the corresponding areas.

Common Misconceptions or Errors

- Students may neglect to add the areas of the decomposed figures to find the total area of the composite figure. Students may also incorrectly add one (or more) of the decomposed figures more than once. To address misconceptions, have students mark or color the figures as they add them to the total to keep track of their work *(MTR.3.1)*.
- Students may not decompose the figure into the most basic figures. To address this misconception, ask students if they can find the area of each of the pieces they have, or if they can break any of them down further to find a more familiar figure *(MTR.5.1)*.

Strategies to Support Tiered Instruction

- Instruction includes writing the area of each decomposed figure inside the original figure and placing a check next to each of the decomposed areas as they are added to determine the total area of the composite figure.
- Teacher provides geometric software for students to interact with composite figures to develop understanding of how to decompose two dimensional figures.
- Teacher provides paper cutouts of different composite figures for students to fold or cut into triangles or quadrilaterals to visually understand how to decompose the area.
- Instruction includes color-coding parallel bases or heights to assist in determining missing measurements of composite figures.
    - For example, given the figure below (assuming the two right triangles have the same side lengths as each other), students can highlight the parallel bases of the rectangle.



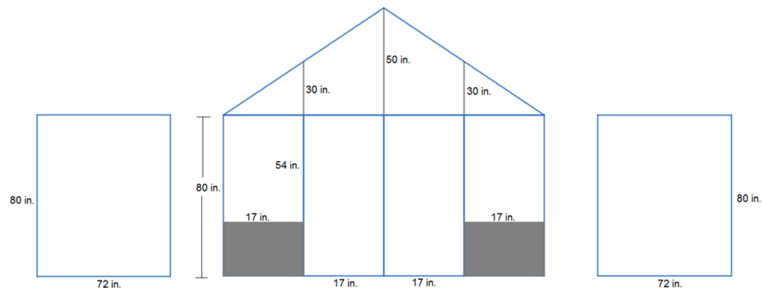
- Teacher has students mark or color the figures as they add them to the total to keep track of their work *(MTR.3.1)*.
- Teacher asks students if they can find the area of each of the pieces they have, or if they can break any of them down further to find a more familiar figure *(MTR.5.1)*.

Instructional Tasks

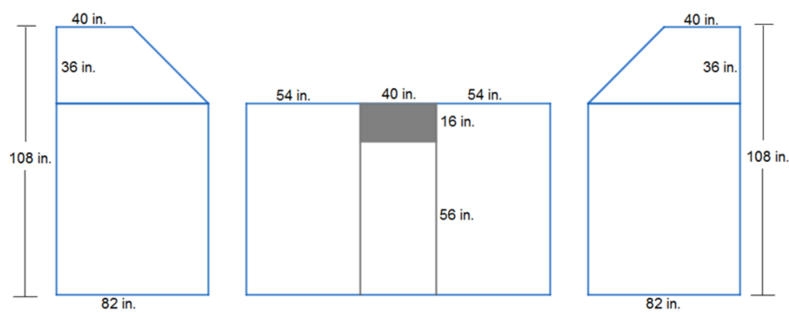
Instructional Task 1 (MTR.6.1)

After a recent storm, Evan has been offered two jobs to replace patio screens. The layouts for the screens needed at both locations are given below. The shaded part represents a stone layout that does not need to be screened.

Job #1



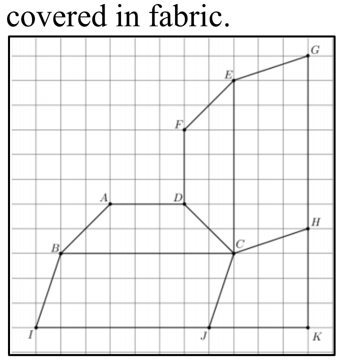
Job #2



If Evan gets paid by the square inch and would like the highest-paying job, which job should he take? Justify your reasoning.

Instructional Task 1 (MTR.3.1)

Tyler and Samantha are building the set for a school play. The design shown below was cut out of wood and now needs to be covered in fabric.



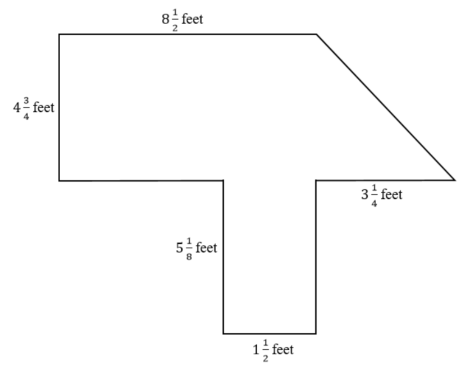
Part A. If each square in the grid has a length of one foot, estimate the total area of wood that needs to be covered. Justify your answer.

Part B. What is the exact total area of the wood, in feet, that needs to be covered? Share your strategy with a partner.

Instructional Items

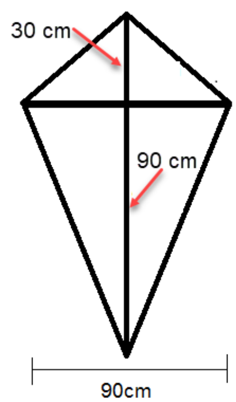
Instructional Item 1

Find the area of the figure below. Note that the figure may not be drawn to scale.



Instructional Item 2

Bena is building a kite based on the design shown below. Determine how much ripstop nylon she will need to purchase for the sail material.



**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*